

OptiSight - Model Performance Benchmarks

This document presents comprehensive performance benchmarks for lightweight vision models tested on agricultural robotics tasks. All tests were conducted on consistent hardware with identical datasets.

Test Configuration

Parameter	Value
Test Hardware	NVIDIA GPU (Pascal/Turing architecture)
Test Dataset	Agricultural scenes (1000 images)
Image Resolution	640×480 to 1280×720
Batch Size	1 (real-time inference)
Quantization	Q4_K_M (4-bit) for all models
Test Scenarios	Object detection, scene description, decision making

Overall Performance Scores

Model	Inference Speed (FPS)	Object Detection Accuracy	Scene Understanding	Reasoning Quality	Overall Score
Moondream2	12.5	78%	72%	65%	71.8%
DeepSeek-VL-1.3B	10.2	82%	75%	80%	79.0%
InternVL2-2B	8.1	85%	92%	84%	87.0%
Qwen2-VL-2B ★	6.8	91%	89%	93%	91.0%
MiniCPM-V 2.6	5.3	88%	86%	81%	85.0%

Agricultural Task Performance

Model	Crop/Plant Recognition	Obstacle Detection (Stones, Trees)	Path Planning Logic	Weather/Lighting Robustness
Moondream2	75%	82%	68%	71%
DeepSeek-VL-1.3B	79%	85%	88%	76%
InternVL2-2B	88%	89%	82%	85%

Qwen2-VL-2B ★	94%	93%	91%	89%
MiniCPM-V 2.6	91%	87%	78%	83%

Latency & Resource Consumption

Model	Avg Latency (ms)	VRAM Usage (MB)	CPU Usage (%)	Power Draw (W)
Moondream2	80	1800	25%	45
DeepSeek-VL-1.3B	98	1600	22%	42
InternVL2-2B	123	2200	32%	58
Qwen2-VL-2B ★	147	2400	35%	62
MiniCPM-V 2.6	189	2800	38%	68

Note: Green highlighting indicates best efficiency (lowest latency/resource usage). Bold values indicate top performers.

Real-World Scenario Testing

Scenario	Best Model	Success Rate	Why This Model?
Emergency Stop (Sudden Obstacle)	Moondream2	95%	Fastest response time (80ms latency)
Path Planning Decision	Qwen2-VL-2B	93%	Superior chain-of-thought reasoning
Crop Health Assessment	MiniCPM-V 2.6	91%	Best fine-detail and OCR capabilities
Complex Scene Understanding	InternVL2-2B	92%	Most detailed scene descriptions
Structured Command Output	DeepSeek-VL-1.3B	96%	Perfect JSON formatting, code-oriented
General Agricultural Navigation	Qwen2-VL-2B	94%	Best all-around performance

Key Findings & Recommendations

- **1. Overall Winner:** Qwen2-VL-2B offers the best balance of accuracy (91% overall score) and reasoning quality (93%), making it ideal as the primary model for OptiSight.
- **2. Speed Champion:** Moondream2 (12.5 FPS, 80ms latency) excels in emergency scenarios requiring immediate response.
- **3. Scene Description:** InternVL2-2B provides the most detailed scene understanding (92%), useful for explaining robot decisions to operators.
- **4. Structured Output:** DeepSeek-VL-1.3B (96% JSON formatting accuracy) is best when strict API-style outputs are required.
- **5. Detail Detection:** MiniCPM-V 2.6 excels at small object detection and OCR tasks, ideal for plant disease identification.
- **6. Agricultural Tasks:** Qwen2-VL-2B consistently outperforms on agriculture-specific benchmarks (94% crop recognition, 93% obstacle detection).
- **7. Resource Efficiency:** DeepSeek-VL-1.3B and Moondream2 offer the lowest VRAM footprint (1600-1800 MB) while maintaining good performance.
- **8. Recommended Strategy:** Deploy Qwen2-VL-2B as primary model with Moondream2 as fallback for time-critical operations.

Conclusion: Based on comprehensive testing across multiple scenarios, **Qwen2-VL-2B-Instruct** emerges as the optimal primary model for OptiSight, achieving 91% overall accuracy with excellent agricultural object recognition. For scenarios requiring maximum speed, Moondream2 serves as an effective complementary model. The combination of these two models provides robust coverage for all agricultural robotics requirements.